

Assignment 11 Solutions

$$12.2 \quad \|\vec{v}_1\| = (4 + 16 + 25)^{1/2} = \sqrt{45} = 3\sqrt{5}$$

$$\|\vec{v}_2\| = (4 + 4 + 4)^{1/2} = \sqrt{12} = 2\sqrt{3}$$

$$\cos \Theta = \frac{\langle \vec{v}_1, \vec{v}_2 \rangle}{\|\vec{v}_1\| \|\vec{v}_2\|} = \frac{(-4 + 8 + 10)}{(3\sqrt{5})(2\sqrt{3})} = \frac{14}{6\sqrt{15}} = \frac{7\sqrt{15}}{45}$$

$$\Theta = \cos^{-1}(7\sqrt{15}/45) \approx 0.924$$

$$\|\vec{v}_1 - \vec{v}_2\| = \|[-4 \quad 2 \quad -3]^T\| = (16 + 4 + 9)^{1/2} = \sqrt{29}$$

$$\frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{3\sqrt{5}} \begin{bmatrix} -2 \\ 4 \\ -5 \end{bmatrix} = \begin{bmatrix} -2/3\sqrt{5} \\ 4/3\sqrt{5} \\ -5/3\sqrt{5} \end{bmatrix} = \begin{bmatrix} -2\sqrt{5}/15 \\ 4\sqrt{5}/15 \\ -\sqrt{5}/3 \end{bmatrix}$$

$$\frac{\vec{v}_2}{\|\vec{v}_2\|} = \frac{1}{2\sqrt{3}} \begin{bmatrix} 2 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/3 \\ \sqrt{3}/3 \\ -\sqrt{3}/3 \end{bmatrix}$$

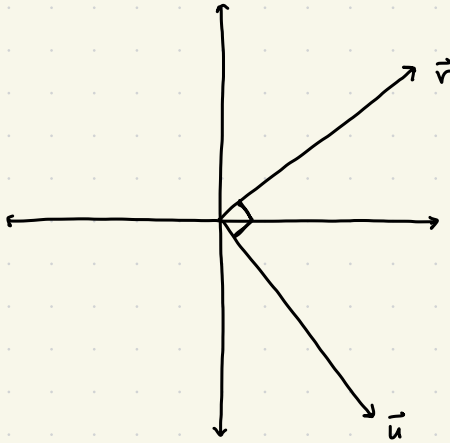
$$12.6 \quad \text{proj}_{\vec{b}_1} \vec{v} = \langle \vec{v}, \vec{b}_1 \rangle \vec{b}_1 = (4\sqrt{2} + 9\sqrt{2}/2) \vec{b}_1 \\ = (17\sqrt{2}/2) \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}^T \\ = \begin{bmatrix} -17/2 \\ 17/2 \end{bmatrix}$$

$$\text{proj}_{\vec{b}_2} \vec{v} = \langle \vec{v}, \vec{b}_2 \rangle \vec{b}_2 = (-4\sqrt{2} + 9\sqrt{2}/2) \vec{b}_2 \\ = (\sqrt{2}/2) \vec{b}_2 \\ = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} 17\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

$$\vec{v} = \text{proj}_{\vec{b}_1} \vec{v} + \text{proj}_{\vec{b}_2} \vec{v} = \left(\frac{17\sqrt{2}}{2}\right) \vec{b}_1 + \left(\frac{\sqrt{2}}{2}\right) \vec{b}_2 \\ = \begin{bmatrix} -17/2 \\ 17/2 \end{bmatrix} + \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$$

$$12.8 \quad \text{proj}_{\vec{u}} \vec{v} = \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{u}, \vec{u} \rangle} \vec{u} = \frac{(63 - 63)}{(\dots)} \vec{u} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$$12.10 \quad \text{proj}_{u_1} \vec{v} = \langle \vec{v}, \vec{u}_1 \rangle \vec{u}_1 = \left(3\sqrt{2}/2 + 6\sqrt{2}/2 \right) \vec{u}_1 \\ = \left(\frac{9\sqrt{2}}{2} \right) \vec{u}_1 \\ = \begin{bmatrix} -9/2 \\ 9/2 \\ 0 \end{bmatrix}$$

$$\text{proj}_{u_2} \vec{v} = \langle \vec{v}, \vec{u}_2 \rangle \vec{u}_2 = \left(-2 + 4 - 2/3 \right) \vec{u}_2 \\ = \left(6/3 - 2/3 \right) \vec{u}_2 \\ = \left(4/3 \right) \vec{u}_2 \\ = \begin{bmatrix} 8/9 \\ 8/9 \\ -4/9 \end{bmatrix}$$

$$\text{proj}_u \vec{v} = \text{proj}_{u_1} \vec{v} + \text{proj}_{u_2} \vec{v} = \begin{bmatrix} -9/2 \\ 9/2 \\ 0 \end{bmatrix} + \begin{bmatrix} 8/9 \\ 8/9 \\ -4/9 \end{bmatrix} = \begin{bmatrix} -81/18 + 16/18 \\ 81/18 + 16/18 \\ 0 + -4/9 \end{bmatrix} = \begin{bmatrix} -65/18 \\ 97/18 \\ -4/9 \end{bmatrix}$$

$$13.1 \quad A^T A = \begin{bmatrix} -3 & 0 & 3 \\ 0 & 3 & -6 \\ 3 & -6 & 3 \end{bmatrix} \begin{bmatrix} -3 & 0 \\ 0 & 3 \\ 3 & -6 \end{bmatrix} = \begin{bmatrix} 18 & -18 \\ -18 & 45 \\ -18 & 45 \end{bmatrix} \begin{bmatrix} 18 & -18 & 18 \\ -18 & 45 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ -6 & 15 & 2 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} -3 & 0 & 3 \\ 0 & 3 & -6 \\ 3 & -6 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 8 \\ 3 \end{bmatrix} = \begin{bmatrix} 18 \\ 6 \\ 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 9 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 8/9 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 17/9 \\ 0 & 1 & 8/9 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 17/9 \\ 8/9 \end{bmatrix}$$

$$13.5 \quad \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 3 & 0 \\ 3 & 0 & 3 \end{bmatrix} \begin{bmatrix} 6 & 3 & 3 & 27 \\ 3 & 3 & 0 & 12 \\ 3 & 0 & 3 & 15 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \\ 4 \\ 7 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 27 \\ 12 \\ 15 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 & 9 \\ 1 & 1 & 0 & 4 \\ 1 & 0 & 1 & 5 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ -1 & 1 & 0 & 4 \\ -2 & 1 & 1 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 5 - x_3$$

$$x_2 = -1 + x_3$$

x_3 is free

12.13 False

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \in \text{Col}(A) \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in \text{Nul}(A) \quad \langle \vec{v}_1, \vec{v}_2 \rangle = 1$$

12.17 True

$$A^{-1} = A^T$$

12.18 False

$$\det(A^T A) = \det(I) = 1 \quad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow \det(A)^2 \Rightarrow \det(A) = \pm 1$$

12.28 True

$$\|A^2 \vec{v}\| = \|A(A\vec{v})\| = \|A\vec{v}\| = \|\vec{v}\|$$

12.29 True

$$\hat{b} = A(A^T A)^{-1} A^T \vec{b} \quad A = \vec{u}$$

$$\begin{aligned} \hat{b} &= \vec{u} (\vec{u}^T \vec{u})^{-1} \vec{u}^T \vec{b} \\ &= u(1)^{-1} \vec{u}^T \vec{b} \\ &= u u^T \vec{b} \end{aligned}$$

13.11 True

$$\text{By def. } \hat{b} \in \text{Col}(A)$$

13.12 False

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \vec{b} = \text{anything}$$

13.20 False

$$A \vec{x} = \vec{b} \text{ has unique LS sol.} \Rightarrow \text{col. of } A \text{ are L.I.}$$

$$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad b = \text{anything}$$

$$12.30 \quad (a) \quad \vec{v}_1 = \begin{bmatrix} -8 \\ 6 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \|\vec{v}_1 - \vec{v}_2\| = \left\| \begin{bmatrix} -12 \\ 5 \end{bmatrix} \right\| = \sqrt{144 + 25} = \sqrt{169} = 13$$

$$(b) \quad \vec{u} = \vec{v}_1 - \vec{v}_2 = \begin{bmatrix} -12 \\ 5 \end{bmatrix} \quad \text{proj}_{\vec{u}} \vec{v}_2 = \frac{\langle \vec{u}, \vec{v}_2 \rangle}{\langle \vec{u}, \vec{u} \rangle} \vec{u} = \frac{-48 + 5}{169} \vec{u} = \frac{-43}{169} \begin{bmatrix} -12 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{516}{169} \\ -\frac{215}{169} \end{bmatrix}$$

$$\vec{v}_3 = \vec{v} + \text{proj}_{\vec{u}} \vec{v}_2 \Rightarrow \vec{v} = \vec{v}_2 - \text{proj}_{\vec{u}} \vec{v}_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{516}{169} \\ -\frac{215}{169} \end{bmatrix} = \begin{bmatrix} \frac{664}{169} \\ \frac{384}{169} \end{bmatrix}$$

$$\|\vec{v}\| = (1 + 9)^{1/2} = \sqrt{10}$$

$$(c) \quad \frac{1}{\sqrt{10}} (4\sqrt{10}) \sqrt{10} = 40$$

$$(d) \quad \begin{bmatrix} 4 & -8 \\ 2 & 6 \end{bmatrix} \quad (e) \quad \det \begin{bmatrix} 4 & -8 \\ 2 & 6 \end{bmatrix} = 24 + 16 = 40$$

$$12.42 \quad (a) \quad \text{proj}_{\vec{v}_1} \vec{v}_2 + \vec{v}_2' = \vec{v}_2 \Rightarrow \vec{v}_2' = \vec{v}_2 - \text{proj}_{\vec{v}_1} \vec{v}_2$$

$$\text{proj}_{\vec{v}_1} \vec{v}_2 = \frac{\langle \vec{v}_1, \vec{v}_2 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1 = \frac{6}{6} \vec{v}_1 = \vec{v}_1$$

$$\vec{v}_2' = \vec{v}_2 - \vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$(b) \quad \vec{v}_3' = \vec{v}_3 - \text{proj}_{\vec{v}_1} \vec{v}_3$$

$$\text{proj}_{\vec{v}_1} \vec{v}_3 = \frac{\langle \vec{v}_1, \vec{v}_3 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1 = \frac{(-8 + 9 + 11)}{6} \vec{v}_1 = 2 \vec{v}_1$$

$$\vec{v}_3' = \vec{v}_3 - 2\vec{v}_1 = \begin{bmatrix} -8 \\ 7 \\ 0 \\ -9 \end{bmatrix}$$

$$(c) \quad \vec{v}_3'' = \vec{v}_3' - \text{proj}_{\vec{v}_2'} \vec{v}_3'$$

$$\text{proj}_{\vec{v}_2'} \vec{v}_3' = \frac{\langle \vec{v}_2', \vec{v}_3' \rangle}{\langle \vec{v}_2', \vec{v}_2' \rangle} \vec{v}_2' = \frac{(-8 - 7 - 9)}{3} \vec{v}_2' = -8 \vec{v}_2'$$

$$\vec{v}_3'' = \vec{v}_3' + 8 \vec{v}_2' = \begin{bmatrix} 0 \\ -1 \\ 0 \\ -1 \end{bmatrix}$$

$$(d) \langle v_1, v_2' \rangle = 2 - 1 - 1 = 0$$

$$\langle v_1, v_3'' \rangle = 0 - 1 + 0 + 1 = 0$$

$$\langle v_2', v_3'' \rangle = 0 + 1 + 0 + -1 = 0$$

$$(e) \text{proj}_{v_1} u = \frac{\langle u, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \frac{7}{6} v_1 = \begin{bmatrix} 7/3 \\ 7/6 \\ 0 \\ -7/6 \end{bmatrix}$$

$$\text{proj}_{v_2'} u = \frac{\langle u, v_2' \rangle}{\langle v_2', v_2' \rangle} v_2' = \frac{-1}{3} v_2' = \begin{bmatrix} -1/3 \\ 1/3 \\ 0 \\ -1/3 \end{bmatrix}$$

$$[u]_{\mathcal{B}} = \begin{bmatrix} 7/6 \\ -1/3 \\ 7/2 \end{bmatrix}$$

$$\text{proj}_{v_3''} u = \frac{\langle u, v_3'' \rangle}{\langle v_3'', v_3'' \rangle} v_3'' = \frac{7}{2} v_3'' = \begin{bmatrix} 0 \\ -7/2 \\ 0 \\ -7/2 \end{bmatrix}$$

13.21 U is orthogonal $\Rightarrow U$ has L.I. columns $\Rightarrow$ L.S. sol is unique

$$\hat{x} = (U^T U)^{-1} U^T b = I^{-1} U^T b = U^T b$$

13.22 The set of L.S. sol. is the solutions of

$C \vec{x} = \hat{b}$ where $\hat{b} = \text{proj}_{\text{col}(C)} b$. This is not a subspace if $\hat{b} \neq \vec{0}$. Let $\hat{x}$ be a L.S. sol.

Then $\{ \hat{x} + \vec{v} : v \in \text{Nul}(C) \}$ is the set of L.S.

sol. There are at most $\dim(\text{Nul}(C)) + 1 = n - k + 1$ L.I. vectors in this set.